

21. Write a Pandas program to swap the cases of a specified character column in a given DataFrame.

Code:

```
import pandas as pd

# Create a sample DataFrame
df = pd.DataFrame({
    'Name': ['John', 'ALICE', 'mArK'],
    'City': ['New York', 'LONDON', 'pArIs']
})

print("Original DataFrame:")
print(df)

# Swap the case of the 'Name' column
df['Name'] = df['Name'].str.swapcase()

print("\nDataFrame after swapping case of 'Name' column:")
print(df)
```

Output:

```

y
Original DataFrame:
   Name      City
0  John  New York
1 ALICE   LONDON
2  mArK   pArIs

DataFrame after swapping case of 'Name' column:
   Name      City
0 jOHN  New York
1 alice   LONDON
2  MaRk   pArIs

```

22. Write a Python program to draw a line with suitable label in the x axis, y axis and a title.

Program:

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 6, 8, 10]
```

```
plt.plot(x, y)
```

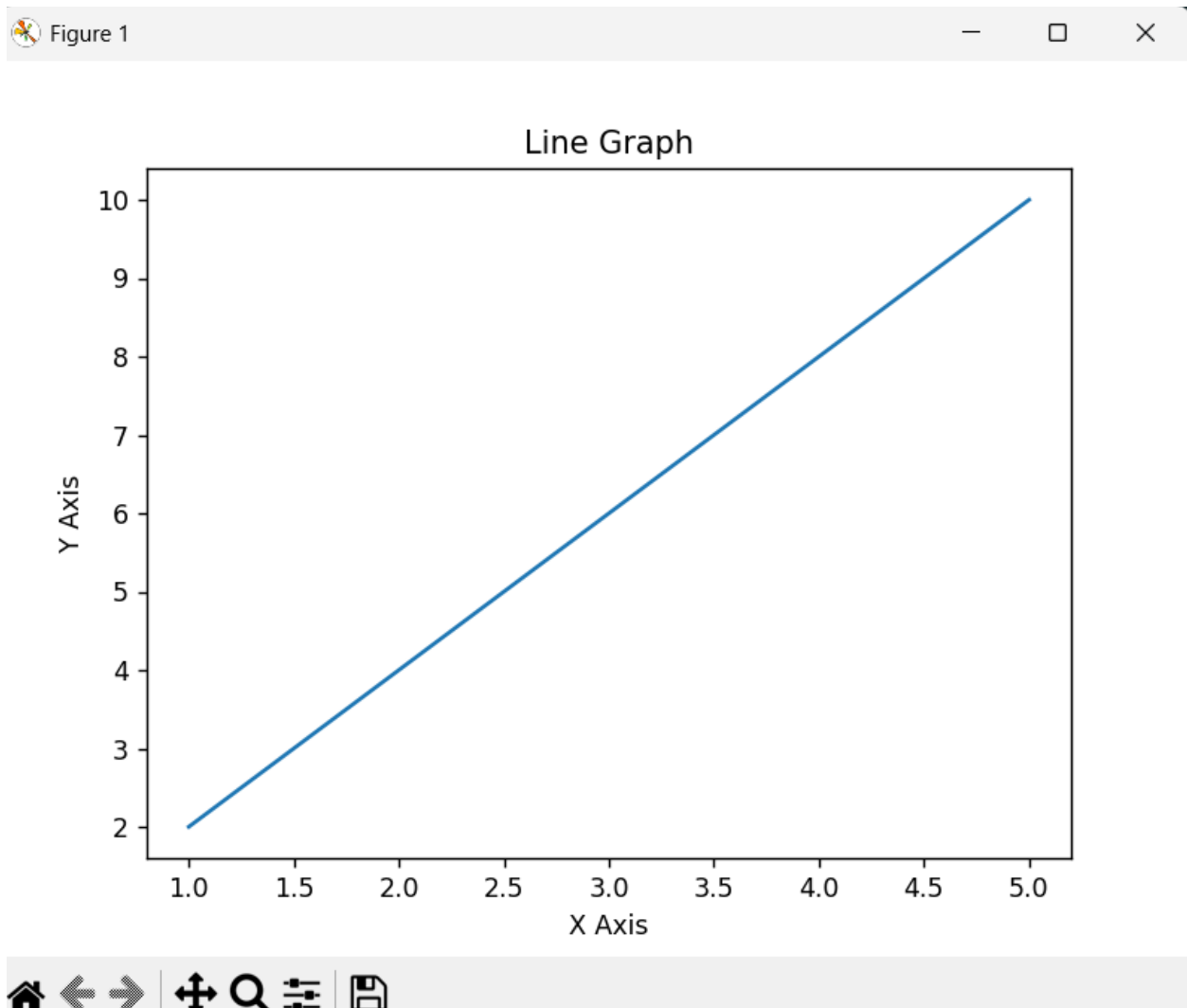
```
plt.xlabel("X Axis")
```

```
plt.ylabel("Y Axis")
```

```
plt.title("Line Graph")
```

```
plt.show()
```

Output:



23. Write a Python program to draw a line using given axis values taken from a text file, with suitable label in the x axis, y axis and a title.

Program :

```
import matplotlib.pyplot as plt
```

```
# Create test.txt

with open("test.txt", "w") as f:

    f.write("1 2\n")

    f.write("2 4\n")

    f.write("3 1\n")


# Read values from test.txt

x = []

y = []


with open("test.txt", "r") as f:

    for line in f:

        a, b = line.split()

        x.append(int(a))

        y.append(int(b))


# Draw line

plt.plot(x, y)


plt.xlabel("x - axis")

plt.ylabel("y - axis")

plt.title("Sample graph")
```

```
plt.show()
```

Output:

